BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 158 · Paper 1: Weakest Topics

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When tracing a loop, should you check its final permitted iteration?

Q2. What is an IDE's editor used for?
A. Fixing physical CPU faults
B. Writing source code
C. Managing user access rights
D. Defragmenting a hard drive

Q3. Draw the standard AND gate symbol, label its input(s) and output, and give one input-output example.

Q4. Either of two sensors can trigger an output. “An OR gate is a rectangle with a memory address inside.” Correct this. Describe the right symbol and why it matters.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Target weak points — Use assessed evidence to select a small number of topics, not indiscriminate revision.

Retrieve independently — Attempt a new problem without notes first.

Fix root causes — Decide whether error arose from knowledge, calculation, interpretation or timing.

Repeat at interval — Reattempt a similar question after a delay to check retention.

New learning

Target weak points: Use assessed evidence to select a small number of topics, not indiscriminate revision.

Retrieve independently: Attempt a new problem without notes first.

Fix root causes: Decide whether error arose from knowledge, calculation, interpretation or timing.

Repeat at interval: Reattempt a similar question after a delay to check retention.

Worked example

An image-size answer forgot to convert bits to bytes. Correct it for 10×20 pixels at 8 bits/pixel.

1. Compute 10×20=200 pixels.

2. Multiply by 8=1600 bits.

3. Divide by 8=200 bytes.

4. Label both quantities and indicate why the original answer was wrong.

We Do · Practise with guidance

Guided 1

A student writes 'WAN is always wireless'. Improve.

Worked method: State correct definition clearly.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the missing operator to convert 968 bits into full bytes.

```
bits = 968
bytes = bits ... 8
print(bytes)
```

write

Question 1

Why might 'faster' alone fail a CPU-performance explanation? [2 marks]

Question 2

A 100×100 image at 1 bit/pixel has how many bits? [2 marks]

Question 3

Explain how spaced retrieval can improve examination readiness. [2 marks]

Exit Ticket · Answer before checking

Identifying the original cause of an error supports targeted revision.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Yes; omitting it can change the final variable values or output.

Q2. B — Writing source code

Q3. Flat input edge; rounded output edge. Usually two inputs and one output. Example: 1 AND 0 = 0.

Q4. Curved input edge; pointed, curved output edge. Usually two inputs. Check the symbol before applying the gate's rule.

I Do / We Do pages · Guided practice

Q1. State correct definition clearly.

You Do pages · Code First

```
bits = 968
bytes = bits DIV 8
print(bytes)
```

Expected result: 121

You Do pages · Independent questions

Q1. It lacks a mechanism such as reduced cache misses or more parallel independent tasks.

Q2. 10,000 bits.

Q3. Repeated recall after intervals helps identify and strengthen knowledge that is not yet secure.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check understanding